Ambient WiFi as a Cheap Radar for Automotive SLAM: Centimetre Localization, a Phantom Ceiling on Mapping, and What That Ceiling Is — in Simulation and on $30 of Silicon

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***Abstract*—We ask whether ambient sub-7 GHz WiFi, received on a moving vehicle and processed with commodity channel state information (CSI), can serve as a low-cost radar/LiDAR replacement for simultaneous localization and mapping (SLAM). Using a physics-based, ray-traced (Sionna RT) substrate and, for the first time in this line of work, a $30 ESP32 hardware testbed, we find a consistent split. *Localization* is centimetre-level and, we show, structurally robust: because the estimator is odometry-anchored and the bistatic CSI measurements are refinement-only, absolute trajectory error is insensitive to access-point–position error up to tens of metres and to sudden, total measurement blackouts. *Mapping*, by contrast, hits a hard ceiling: with realistic commodity CSI, most detections are phantoms ($\approx$89%) accompanied by a several-metre range bias — a front-end and geometry limit, not a path-discrimination or resolution limit, and one a learned filter cannot repair. Comparing the same substrate against real LiDAR (KITTI) and a simulated 77 GHz FMCW radar shows the ceiling is neither carrier-specific nor WiFi-specific: the radio carrier does not matter, geometry does (a monostatic on-vehicle transmitter nearly eliminates the phantoms), and wider bandwidth makes them worse. Finally, we reproduce the delay-domain reading on real ESP32 silicon. The practical message: WiFi localization rivals far costlier sensors, while WiFi mapping is capped by a front-end/geometry phantom ceiling that we characterize and, in part, show how to escape.**

***Index Terms*—WiFi sensing, channel state information, SLAM, passive radar, integrated sensing and communication, automotive, ESP32, low-cost hardware, multipath.**

TABLE I
THE MAPPING CEILING DECOMPOSES INTO THREE MECHANISMS; THE SMALLEST IS THE ONE PAPER 1 NAMED. (CONTROLLED / STREET)

Mechanism	Magnitude	Role
Phantom detections	89.2%/89.5%	**dominant** (no real path)
Estimation range bias	6.45 m (controlled)	ruins correct paths
Path discrimination	2–8%	real, but *smallest*

I. Introduction

Contributions

- A physics-based feasibility study of on-vehicle, mobile, outdoor WiFi as a SLAM front-end — an unoccupied point in the design space (prior passive-WiFi-radar is indoor/static/infrastructure-fixed; the one moving-platform passive radar uses 5G).
- Identification and mechanism of the *mapping phantom ceiling* ($\approx$89% phantom detections + range bias), and proof it is not a discrimination or resolution limit and not repairable by a learned filter.
- Evidence the ceiling is *universal*: carrier-independent and not WiFi-specific, via a controlled comparison against LiDAR and a 77 GHz radar on one substrate; geometry (monostatic vs bistatic), not carrier or bandwidth, is decisive.
- A robustness characterization: localization is structurally tolerant of access-point–position error and measurement blackouts (Sec. V-B).
- The first delay-resolved reflector reading from an ESP32 on $30 of hardware, moving the claim off simulation (Sec. VI).

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TABLE II
ABLATION ON ONE SUBSTRATE (CONTROLLED_WALL, 3 SEEDS). GEOMETRY, NOT CARRIER OR BANDWIDTH, SETS THE PHANTOM RATE.

Cell	Front-end / geometry	Carrier/BW	Phantom rate
M	MUSIC, bistatic	5.2 GHz/160 MHz	— (IoU 0.003)
A	CFAR, bistatic	5.2 GHz/160 MHz	$18.2 \pm 0.6\%$
B	CFAR, monostatic	5.2 GHz/160 MHz	$\mathbf{0.1 \pm 0.2}\%$
C	CFAR, monostatic	77 GHz/160 MHz	0.0%
D	CFAR, monostatic	77 GHz/**4 GHz**	$9.0 \pm 1.5\%$

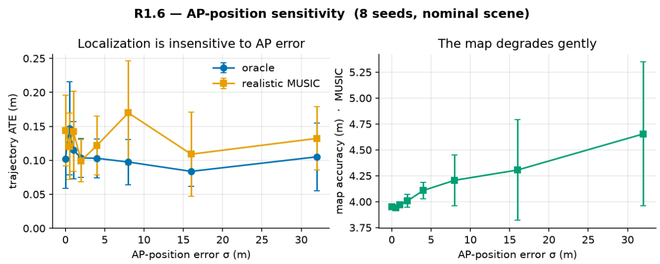

Fig. 1. [R1.6] Trajectory ATE is insensitive to AP-position error to 32 m (structural: odometry-anchored, refinement-only measurements); the map degrades gently and monotonically.

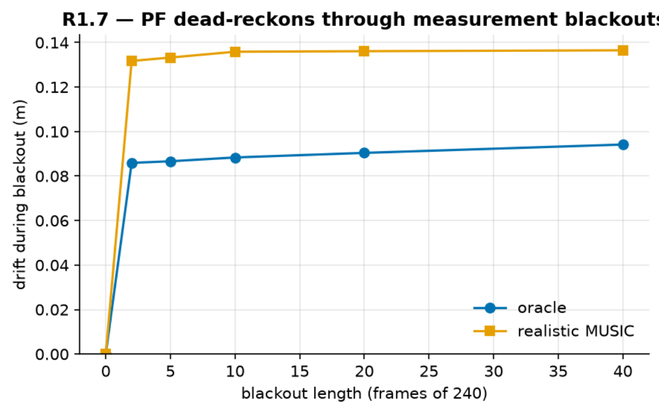

Fig. 2. [R1.7] The particle filter dead-reckons through a total measurement blackout: a 40-frame (17% of run) outage adds ≈0.1 m drift and re-locks immediately.

II. RELATED WORK

III. METHOD: ONE SUBSTRATE, THREE SENSORS, ONE TESTBED

A. Ray-traced channel and scenes

B. CSI front-end: joint 2-D (delay–angle) MUSIC

C. Bistatic-ellipse mapping and particle-filter SLAM

D. Complexity and real-time feasibility

E. LiDAR and 77 GHz radar baselines

F. ESP32-S3 hardware testbed

IV. FEASIBILITY: LOCALIZATION WORKS, MAPPING IS TWO-TIER

V. THE MAPPING CEILING AND ITS MECHANISM

A. Universality: WiFi vs 77 GHz radar

B. Robustness to AP-position error and measurement loss

VI. HARDWARE VALIDATION ON $30 OF SILICON

VII. DISCUSSION

VIII. CONCLUSION

REPRODUCIBILITY

All code, configs, and result data are in the repository above (AGPL-3.0); the simulation is Sionna RT; the hardware pipeline and firmware are included.